

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength of a particle is given by the formula $\lambda = h / \sqrt{2 m E}$, where λ is the wavelength, h is the Planck's constant, m is the mass of the particle, and E is its kinetic energy. The kinetic energy of an electron is given as 1 eV. We need to convert this energy into joules, which is $1 \text{ J} = 1.6 \times 10^{-19} \text{ J}$ and $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. The mass of an electron is approximately $9.11 \times 10^{-31} \text{ kg}$. Now, we can use the formula to calculate the de Broglie wavelength. First, we'll calculate the speed of the electron using the kinetic energy, then multiply that speed